

✓ Most Repeated Questions / Topics

1. Boolean Algebra & K-Map Simplification

- Simplify Boolean functions using K-map (SOP form)
- Draw logic diagrams using AND, OR, NOT
- Frequently appears in **Q1**

2. Instruction Cycle / Micro-operations

- Fetch cycle micro-operations
- Sequence of fetching instruction (involving PC, MAR, IR, DR, AC)
- Appears every year

3. Cache Memory

- Need and advantages
- Direct mapping
- Two-way set associative mapping
- With diagrams and examples

4. Von Neumann Architecture

- Explained with diagram
- Difference from Harvard architecture occasionally asked

5. 8086 Assembly Language Programs

- Convert ASCII to binary
- Find smallest/largest of numbers
- Move data, loop, or compare values using registers
- Use of `INT 21h`

6. Addressing Modes

- Immediate, Direct, Indirect, Indexed, Register
- Explanation with examples
- Asked as 4- or 5-mark question

7. DMA (Direct Memory Access)

- Block diagram
- Functions and need
- Comparison with programmed I/O and interrupt-driven I/O

8. Flip-Flops

- J-K, D flip-flop: logic diagrams, characteristic table, excitation table

9. I/O Techniques

- Programmed I/O
- Interrupt-driven I/O
- Differences from DMA
- Use of flowcharts/diagrams

10. Floating Point and Number Representations

- Convert binary, octal, hex
- 1's and 2's complement
- Floating point format (normalized, biased exponent)

11. 8086 Instruction Explanation

- Instructions like XCHG, MUL, ADD, SHL, SHR, LEA, TEST, ROL, RCL, ADC, XOR, INT
- Their syntax, function, and example

12. Segment Registers and Physical Address Calculation

- CS, DS, SS, SP, IP usage to calculate 20-bit address
- Often tested with given values

★ Other Important Topics That Frequently Appear

- Interrupts: vector table, handling
- Control Unit: Wilkes Control, Microprogrammed Control Unit
- Role of Stack in CALL/RET
- Memory hierarchy: RAM types, ROM vs Flash, RAID
- Registers and their operations (e.g., addition, AND, complement)
- Multiplexers (2x1, 4x1) and their logic diagrams
- Accumulator-based architecture: expressions like $A = B * C + D$
- Memory chips and address line calculation
- Even parity and error detection

✦ Summary Flash Note for Preparation Strategy

Topic	Priority	Type
Boolean simplification using K-map	★★★★	Must Practice

Topic	Priority	Type
Instruction cycle & micro-ops	★ ★ ★ ★	Conceptual
Cache memory (direct & associative)	★ ★ ★ ★	Theory + Diagram
Assembly programs in 8086	★ ★ ★ ★	Hands-on
Addressing modes	★ ★ ★ ★	Theoretical
DMA vs I/O techniques	★ ★ ★ ★	Comparative
Flip-Flops	★ ★ ★	Logic Design
Von Neumann architecture	★ ★ ★	Diagram Based
Segment & offset address calculation	★ ★ ★	Numerical
Floating point conversion	★ ★ ★	Numerical